

Decoding Longevity: A Clustering Analysis of Health and Lifestyle Indicators

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Abstract—Understanding how lifestyle choices impact health and longevity is crucial for improving public health strategies. This study applies machine learning clustering techniques to analyze lifestyle patterns and their influence on health risks using the Diabetes Health Indicators Dataset. K-Means, Agglomerative Clustering, and DBSCAN were used to identify six distinct health clusters. The results showed that smoking, high cholesterol, poor mental health, and financial struggles are key risk factors. K-Means produced the most well-defined clusters, while Agglomerative Clustering showed more overlap, and DBSCAN failed to form enough clusters for a radar chart. These insights can help healthcare providers and policymakers develop targeted interventions to reduce chronic disease risks. The study's limitations include dataset biases and computational constraints, which affected clustering performance. Future research should explore additional health factors and improved clustering techniques. This study highlights how data-driven clustering can identify high-risk groups, supporting early interventions and better public health outcomes.

Index Terms—Clustering, K-Means, Machine Learning, Longevity, Health Indicators, Public Health, Lifestyle Analysis, Diabetes, Principal Component Analysis, DBSCAN, Agglomerative Clustering, Chronic Diseases, Risk Factors, Health Data Analytics.

I. INTRODUCTION

A person's lifestyle is indicative of whether they live longer or die prematurely. A person with a healthy lifestyle is

expected to live longer than someone who has an objectively unhealthy lifestyle. A comprehensive study published in *Circulation* analyzed over 42,000 individuals and found that adopting five low-risk lifestyle factors—healthy diet, regular exercise, healthy weight, no smoking, and moderate alcohol intake—can increase life expectancy at age 50 by 14.0 years for women and 12.2 years for men [4].

Living a healthy lifestyle, being fit, and having a balanced diet are all factors to be considered when trying to decipher one's life expectancy. This study aims to decode how long a person lives through clustering algorithms and analysis of behavioral patterns and trends. The research will involve experimenting on a Diabetes Health Indicators Dataset from Kaggle, clustering different groups of people who have distinct lifestyle patterns, and profiling each of these clusters into different age brackets with the intention of discovering which factors contribute to a higher risk of reduced life expectancy.

Life expectancy can be influenced by various lifestyle factors, but accurately predicting how long a person may live based on their lifestyle habits can be a challenge. Understanding which behavioral patterns contribute to a longer or shorter lifespan can help in designing effective health interventions. However, current models may not fully capture the complexity of lifestyle factors, particularly in relation to conditions like diabetes.

This study proposes the use of clustering algorithms to analyze behavioral trends and lifestyle patterns in individuals from the **Diabetes Health Indicators Dataset** from Kaggle. By grouping individuals with similar health habits, the analysis aims to identify key risk factors that

contribute to reduced life expectancy. This will help in categorizing individuals based on their likelihood of premature mortality and providing targeted health recommendations.

Understanding the relationship between lifestyle choices and life expectancy is crucial for public health planning and disease prevention. By identifying high-risk groups and key contributing factors, this study can help policymakers, healthcare providers, and individuals make informed decisions to improve health outcomes and reduce the burden of chronic diseases like diabetes.

Individuals impacted by this problem include those at risk of lifestyle-related diseases, such as diabetes and heart disease, as well as healthcare professionals seeking data-driven insights for patient care. The potential benefactors of this solution include medical researchers studying the effects of lifestyle on longevity and individuals aiming to assess and improve their own lifestyle choices and health outcomes. This solution can also aid future researchers conducting lifestyle and longevity studies.

II. REVIEW OF RELATED LITERATURE

Diet, exercise, and weight control are important lifestyle variables that contribute to the development and treatment of type 2 diabetes mellitus (T2DM), which has a substantial influence on longevity.

A person's lifetime is greatly impacted by lifestyle variables including food, weight control, and physical exercise, which are critical in the development and treatment of type 2 diabetes. People with poor glucose tolerance may lower their risk of type 2 diabetes by around 58% by keeping a balanced diet and getting frequent exercise. On the other hand, a lower life expectancy is linked to an earlier diagnosis of diabetes; for example, a diagnosis at age 30 may shorten life expectancy by as much as 14 years.[1]

To create successful lifestyle treatments for the control of diabetes, a number of theoretical models have been used. The Transtheoretical Model (TTM), which measures a person's willingness to adopt new, healthier habits, has been used to encourage dietary and physical activity modifications in people with diabetes and prediabetes.[2]

Over the years, a great deal of research has been done on the connection between lifestyle variables and lifespan, which has provided important new information on how activities affect longevity. Three important studies that illustrate the development of this area from various eras to the present are listed below:

Study of Seven Countries (Started in 1956)

This ground-breaking epidemiological research, which was led by Ancel Keys, methodically investigated the connections between nutrition, lifestyle, coronary heart disease, and stroke in various communities throughout the globe. It laid the groundwork for our knowledge of how lifestyle choices affect longevity by being the first to draw attention to the relationship between heart disease and dietary patterns and lifestyle.

The Health Study of Nurses (Started in 1976)

The health of women was the main focus of this long-term research, which gathered information on a range of lifestyle variables, such as smoking, physical activity, and food. The results highlight the significance of healthy habits in prolonging lifespan and have broadened our knowledge of how these variables affect the risk of chronic illnesses and total mortality.

Current Research on Life Expectancy and Healthy Lifestyle Factors (2024)

In Taiwan, a national cohort study examined the effects of five healthy lifestyle choices on life expectancy and health care costs: abstaining from smoking, limiting alcohol intake, exercising regularly, eating enough fruits and vegetables, and keeping a healthy weight. According to the research, following these habits was linked to a significant rise in life expectancy and a decrease in medical expenses, highlighting the crucial role that lifestyle plays in determining longevity.[3]

Risk for Diabetes and Chronotype

People who enjoy late bedtimes (often known as "night owls") are 46% more likely to acquire type 2 diabetes than people who wake early, according to a research with over 5,000 participants. There are many reasons for this elevated risk, including:

- increased risk of obesity.
- unhealthy eating patterns.
- increased usage of tobacco.[5]

Type 2 Diabetes at an Early Stage and the Risk of Death

According to research from the University of Oxford, those who are diagnosed with type 2 diabetes before the age of 40 have a roughly four-fold increased risk of dying young when compared to the general population. This emphasizes the vital need of:

- early detection of those who are at danger.
- therapy plans specifically designed for younger folks.
- The risk of type 2 diabetes may be decreased by taking preventive actions including eating a balanced diet and exercising often.
- These studies demonstrate the significant influence of lifestyle variables, especially physical activity and

sleep habits, on the development of type 2 diabetes, as well as on quality of life and mortality risk after diagnosis. Improving outcomes for those with diabetes or at risk for the disease requires addressing these characteristics via treatment and preventative strategies.[6]

III. METHODOLOGY

A. Data Collection

The researchers used the “Diabetes Health Indicators Dataset” from Kaggle created by a Data Scientist named Alex Teboul. The original uncleaned data came from the CDC - 2015 BRFSS Survey Data which is a Behavioral Risk Factor Surveillance System that collect state data about U.S residents regarding their health-related risk behaviors.

The researchers utilized the 50/50 split version as the original dataset is highly biased towards one side and is not balanced. The data is clean with 70,692 survey responses and is structured in binary form. The data is labeled as such:

Diabetes_binary: 0 = no diabetes 1 = diabetes & prediabetes.

HighBP 0: = no high BP 1 = high BP.

HighChol: 0 = no high cholesterol 1 = high cholesterol.

CholCheck: 0 = no cholesterol check in 5 years 1 = yes cholesterol check in 5 years

BMI: Body Mass Index

Smoker: Have you smoked at least 100 cigarettes in your entire life? [Note: 5 packs = 100 cigarettes] 0 = no 1 = yes

Stroke: (Ever told) you had a stroke. 0 = no 1 = yes

HeartDiseaseorAttack: coronary heart disease (CHD) or myocardial infarction (MI) 0 = no 1 = yes

PhysActivity: physical activity in past 30 days - not including job 0 = no 1 = yes

Fruits: Consume Fruit 1 or more times per day 0 = no 1 = yes

Veggies: Consume Vegetables 1 or more times per day 0 = no 1 = yes

HvyAlcoholConsump: (adult men >=14 drinks per week and adult women >=7 drinks per week) 0 = no 1 = yes

AnyHealthcare: Have any kind of health care coverage, including health insurance, prepaid plans such as HMO, etc. 0 = no 1 = yes

NoDocbcCost: Was there a time in the past 12 months when you needed to see a doctor but could not because of cost? 0 = no 1 = yes

GenHlth: Would you say that in general your health is: scale 1-5 1 = excellent 2 = very good 3 = good 4 = fair 5 = poor

MentHlth: days of poor mental health scale 1-30 days

PhysHlth: physical illness or injury days in past 30 days scale 1-30

DiffWalk: Do you have serious difficulty walking or climbing stairs? 0 = no 1 = yes

Sex: 0 = female 1 = male

Age: 13-level age category (_AGEG5YR see codebook) 1 = 18-24 9 = 60-64 13 = 80 or older

Education: Education level (EDUCA see codebook) scale 1-6 1 = Never attended school or only kindergarten 2 = elementary etc.

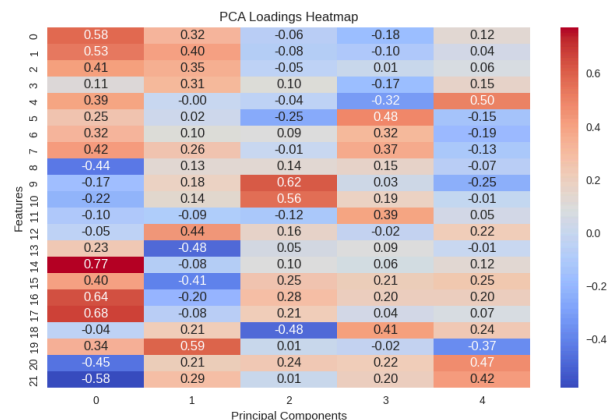
Income: Income scale (INCOME2 see codebook) scale 1-8 1 = less than \$10,000 5 = less than \$35,000 8 = \$75,000 or more

B. Data Pre-Processing

The researchers decided that cleaning the data was unnecessary as the data provided was already cleaned and classified into binary classifications.

The researchers used PCA Loadings Heatmap for feature selection in which the higher the value of the feature, the more the feature influences the principal component. The researchers chose those with positive values and excluded the features with negative values as shown in the figure. The features left are the following:

Diabetes_binary, HighBP, HighChol, BMI, Smoker, Stroke, HeartDiseaseorAttack, NoDocbcCost, GenHlth, MentHlth, PhysHlth, DiffWalk, Age



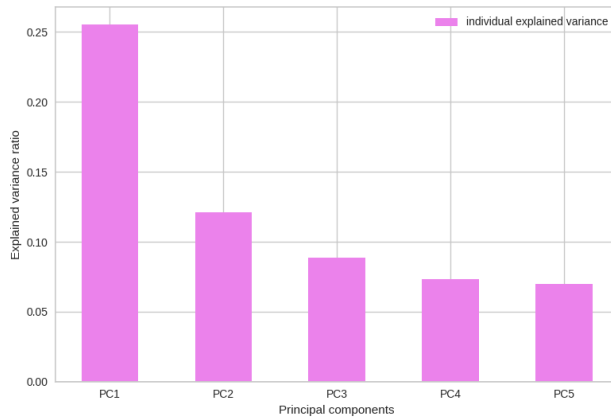
The researchers used a StandardScaler as the scaling method gave the best clustering performance compared to the MinMaxScaler and without scaling. However, scaling the data decreased the variance of the principal components throughout the scree plot.

	Diabetes_binary	HighBP	HighChol	BMI	Smoker	Stroke	\
0	0.0	1.0	0.0	26.0	0.0	0.0	
1	0.0	1.0	1.0	26.0	1.0	1.0	
2	0.0	0.0	0.0	26.0	0.0	0.0	
3	0.0	1.0	1.0	28.0	1.0	0.0	
4	0.0	0.0	0.0	29.0	1.0	0.0	
...	...	...	...	...	...	...	
70687	1.0	0.0	1.0	37.0	0.0	0.0	
70688	1.0	0.0	1.0	29.0	1.0	0.0	
70689	1.0	1.0	1.0	25.0	0.0	0.0	
70690	1.0	1.0	1.0	18.0	0.0	0.0	
70691	1.0	1.0	1.0	25.0	0.0	0.0	

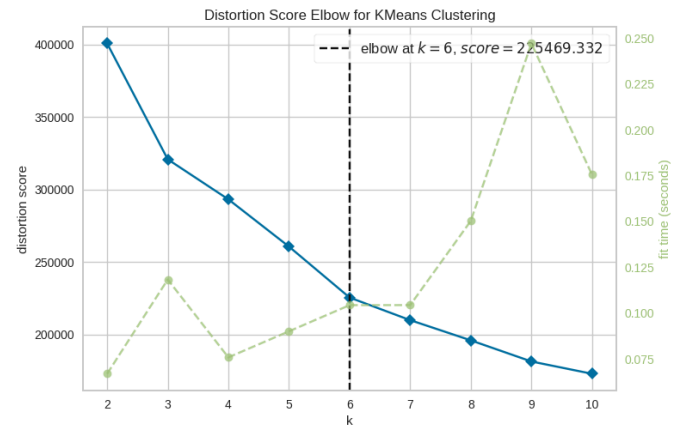
	HeartDiseaseorAttack	NoDocbcCost	GenHlth	MentHlth	PhysHlth	\
0	0.0	0.0	3.0	5.0	30.0	
1	0.0	0.0	3.0	0.0	0.0	
2	0.0	0.0	1.0	0.0	10.0	
3	0.0	0.0	3.0	0.0	3.0	
4	0.0	0.0	2.0	0.0	0.0	
...	...	...	...	...	...	
70687	0.0	0.0	4.0	0.0	0.0	
70688	1.0	0.0	2.0	0.0	0.0	
70689	1.0	0.0	5.0	15.0	0.0	
70690	0.0	0.0	4.0	0.0	0.0	
70691	1.0	0.0	2.0	0.0	0.0	

	DiffWalk	Age	\
0	0.0	4.0	
1	0.0	12.0	
2	0.0	13.0	
3	0.0	11.0	
4	0.0	8.0	
...	...	...	
70687	0.0	6.0	
70688	1.0	10.0	
70689	1.0	13.0	
70690	1.0	11.0	
70691	0.0	9.0	

The researchers also used the Principal Component Analysis (PCA) to reduce the number of features but still preserve an acceptable total variance. The number of Principal Components used is **5 components** with a **combined variance of 0.606** as shown in the scree plot figure below.



The researchers also utilized the elbow method to determine the optimal number of clusters. The elbow method revealed that the optimal number of clusters is 6 and the distortion score is 225,469 as show in the figure below.



C. Experimental Setup

The following tools and libraries used are as follows: Pandas, Seaborn, Matplotlib, Sklearn, Numpy, Yellowbrick, Plotly.

For the Agglomerative Clustering and DB Scan Algorithm, the researchers fractioned out the data into 30 percent, as the devices used were not able to handle the memory demand of the algorithms.

For the Agglomerative Clustering, the parameters used are: n_clusters = 6, metric = euclidean, linkage = ward. For the DBScan, the parameters used are: epsilon = 2, minimum samples = 10, metric = euclidean.

D. Algorithm

In this study, the researchers conducted this study using unsupervised machine learning algorithms to analyze the and cluster the dataset effectively. They implemented Principal Component Analysis (PCA) to reduce dimensionality while preserving essential variance, ensuring that the dataset retained meaningful information while improving computational efficiency. To achieve optimal clustering, they employed three distinct clustering algorithms: K-Means Clustering, Agglomerative Hierarchical Clustering, and DBSCAN (Density-Based Spatial Clustering of Applications with Noise).

The choice of these specific algorithms was based on their individual strengths in handling different types of data structures. PCA was utilized to reduce feature redundancy and improve clustering performance by transforming the dataset into a lower dimensional space. K-Means Clustering was used to provide a more flexible, hierarchical view of cluster relationships without requiring a predefined number of clusters. Additionally, DBSCAN was incorporated to detect clusters of varying densities and handle the noise and outliers more effectively than centroid-based clustering methods.

To optimize the performance of these models, the researchers applied several techniques. For PCA, they analyzed the explained variance ratio to determine the number of the principal components that capture the most variance in the dataset.

```

Explained variance of PC1 0.2551462622376482
Explained variance of PC2 0.12124680305825795
Explained variance of PC3 0.08875426723423789
Explained variance of PC4 0.07351679106347365
Explained variance of PC5 0.06995241344079989

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The elbow method was employed in K-Means clustering to determine the optimal number of clusters. Furthermore, k-means++ initialization was used to improve the convergence of the algorithm. In Agglomerative Clustering, the Ward's linkage method and Euclidean distance metric were chosen to merge clusters optimally. For DBSCAN, the researchers fine-tuned parameters such as the epsilon (eps) value and minimum samples per cluster to enhance cluster detection. Finally, silhouette scores were computed to assess and compare the performance of each clustering algorithm.

E. Evaluation Metrics

The researchers used silhouette score for the 3 clustering algorithms to measure the clustering performance of each algorithm. The achieved silhouette scores are shown below in the figure

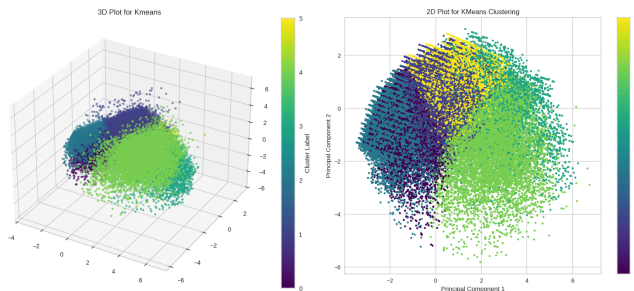
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K means Silhouette Score: 0.2310054243129038
Agglomerative Silhouette Score: 0.1825845245007552
DBScan Silhouette Score: 0.5122719350922502

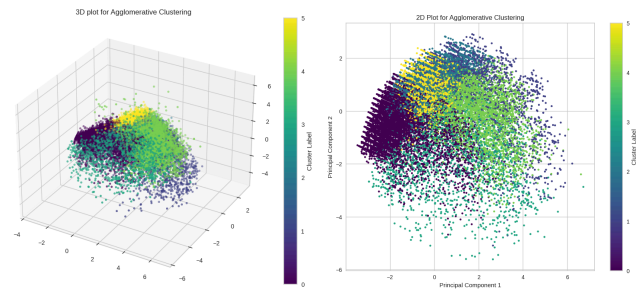
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F. Comparison of Clustering Algorithms

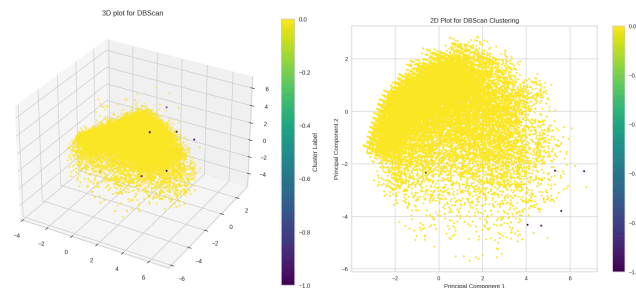
The researchers used 3 clustering algorithms. The following algorithms used are K-means, Agglomerative and DBScan clustering. We compared the three algorithms to each other and analyzed their clustering performance by plotting the first 3 principal components in a 3D scatter plot to visually present the clusters. We also used a 2D plot using the first 2 principal components to visualize the clusters.



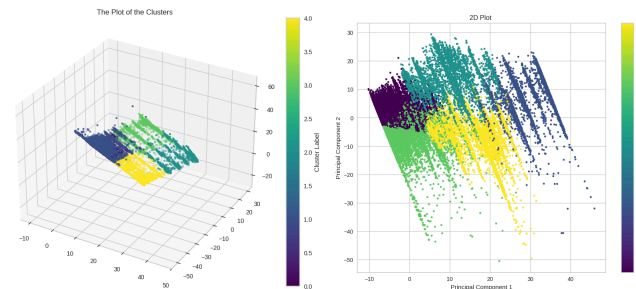
2D and 3D plot for K-means



2D and 3D plot for Agglomerative

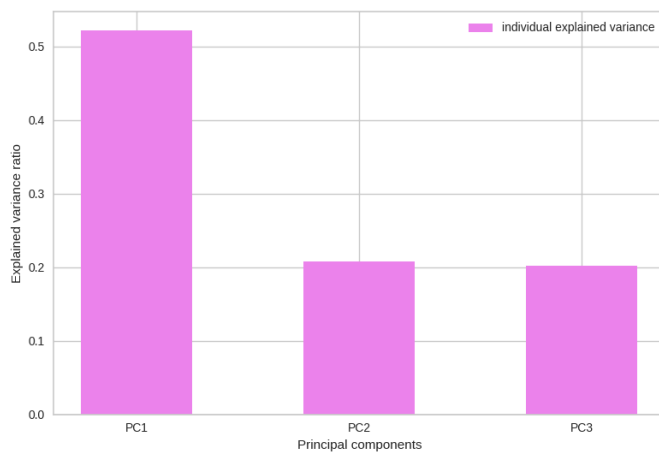


2D and 3D plot for DBScan



2D and 3D plot for K-means without scaling method

Comparing the scaled and unscaled data, the scaled data showed better clustering visually. Despite this, the unscaled data had more combined total variance across the principal components. While the scaled data showed better clustering visually, the total variance only amounts to 0.60 while having 5 principal components. This shows that scaled data had better clustering visually, but had lesser variance while the unscaled data had a higher combined variance and had worse clustering performance, as the clusters looked flat and did not show well defined clusters as compared to the scaled data.



Scree plot for unscaled data

IV. RESULTS AND DISCUSSION

The research successfully identified behavioral patterns linked to reduced life expectancy by applying Principal Component Analysis (PCA) and K-Means clustering. Instead of focusing solely on diabetes, the study aimed to uncover lifestyle-based clusters that highlight risk factors such as smoking, physical inactivity, poor diet, and excessive alcohol consumption.

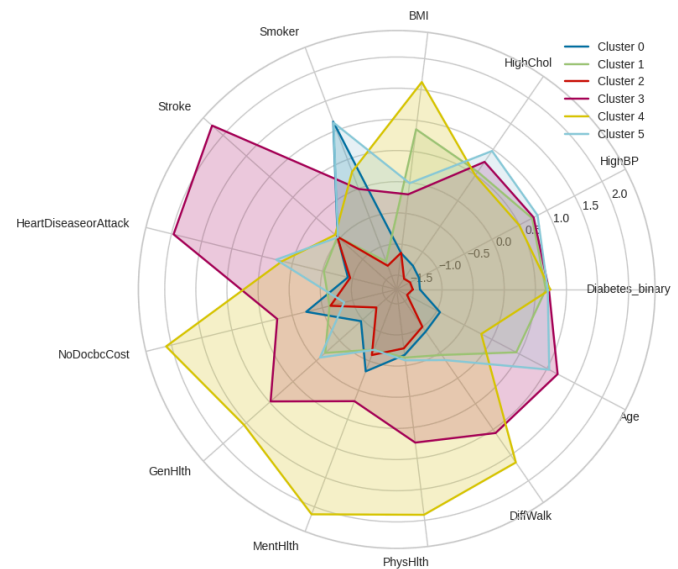
The Elbow Method determined the optimal number of clusters, and the results demonstrated that PCA improved cluster separability by reducing redundant information. The findings revealed distinct groups with varying health behaviors, emphasizing the need for targeted public health interventions to address modifiable risk factors.

The clustering approach provided clear trends, showing that individuals with sedentary lifestyles, unhealthy diets, and high alcohol consumption fell into clusters associated with lower life expectancy, while those with active and healthier habits were linked to better health outcomes. The study's results aligned with existing research, confirming that lifestyle choices significantly impact longevity. While the Elbow Method was initially used to determine the optimal number of clusters, further evaluation using Silhouette Scores provided a deeper understanding of clustering quality.

The analysis showed that DBSCAN achieved the highest silhouette score (0.5123), suggesting well-defined clusters, followed by K-Means (0.2310) and Agglomerative Clustering (0.1826). These results indicate that DBSCAN may be better suited for identifying behavioral risk groups, as it handles irregularly shaped clusters and noise more effectively than the other methods.

From the Radar Charts that the researchers produced in K-means and Agglomerative Clusters they observed the

following key lifestyle/behavioral patterns from the clusters that contribute to a higher risk of reduced life expectancy:



Radar Chart for K-means Clustering

Key observations for K-Means Clustering:

Cluster 0: This group represents younger individuals who smoke frequently but do not yet have chronic conditions like diabetes, high cholesterol, or high blood pressure. However, their physical health is already poor, possibly due to unhealthy habits. They may also face some healthcare access issues.

Cluster 1: This cluster represents middle-aged individuals with moderate health risks. They do not smoke, but they have higher cholesterol, blood pressure, and diabetes prevalence compared to Cluster 0. Their physical and mental health are poor, and they may have some difficulty accessing healthcare.

Cluster 2: This group represents the healthiest individuals with the lowest chronic disease risks. They are likely young adults with no major health conditions, low BMI, and low cholesterol and blood pressure. However, their mental and physical health are still poor, indicating potential lifestyle stressors or early-stage health problems.

Cluster 3: This cluster consists of older individuals with high chronic disease risks, including heart disease, stroke, and diabetes. They experience high difficulty walking, likely due to age-related health decline. Their mental and physical health are moderate, but they face some financial challenges.

Cluster 4: This cluster shows that they struggle severely with health and financial issues. They have high BMI, difficulty walking, extremely poor physical and mental health, and very limited access to healthcare. However, they do not have high heart disease or stroke prevalence yet, but they are at risk.

Cluster 5: This represents middle-aged individuals with significant health risks. They smoke frequently, have high cholesterol and blood pressure, and are at risk for diabetes.

They do not yet have high heart disease rates, but they are likely to develop chronic illnesses if their lifestyle does not change.

In summary for K-means Clustering, Clusters 2 and 0 are the healthiest, with Cluster 2 being the lowest risk group overall. Clusters 3, 4, and 5 represent populations with increasing health risks, with Cluster 4 facing the most severe physical, mental, and financial challenges. Cluster 5 is at high risk due to lifestyle factors, while Cluster 3 has already developed multiple chronic diseases.

This group may include older adults or those with mobility issues and chronic diseases.

Cluster 5: Moderate values across most health indicators, with no extreme spikes. May represent a middle-ground population with manageable health risks. Shows slightly lower rates of NoDocbcCost, meaning better access to healthcare than Cluster 3.

In summary for Agglomerative Clustering, Clusters 1 & 4 represent high-risk groups for chronic diseases and mobility issues. Cluster 2 highlights smoking and BMI concerns. Cluster 3 faces significant financial and mental health challenges. Clusters 0 & 5 have relatively better health outcomes.

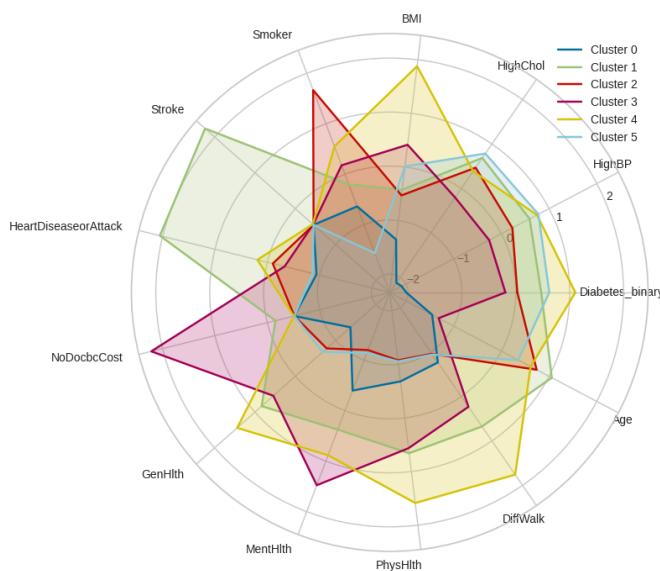
V. CONCLUSION

This study explored the relationship between lifestyle behaviors and longevity using clustering techniques. The goal was to identify distinct health risk groups and analyze key factors influencing life expectancy. By applying K-Means, Agglomerative Clustering, and DBSCAN, six health clusters were identified, with Cluster 2 representing the healthiest individuals, while Clusters 3, 4, and 5 showed increasing risks, with Cluster 4 being the most vulnerable due to severe health and financial struggles.

The findings highlighted that smoking, high cholesterol, poor mental health, and financial struggles significantly impact health outcomes. Among the models tested, K-Means produced the most distinct and well-defined clusters, making it the most effective at identifying individuals with similar lifestyle risk factors. The Agglomerative Clustering model produced looser, more overlapping clusters, making it less effective in separating distinct health groups. DBSCAN struggled to form meaningful clusters, resulting in an inability to generate a radar chart due to insufficient cluster formation. This limitation suggests that DBSCAN may not be ideal for datasets with overlapping health patterns, where clear group distinctions are necessary.

These insights can be valuable for healthcare providers, policymakers, and researchers looking to design targeted interventions for individuals at higher risk of chronic diseases. By understanding which lifestyle factors contribute most to health decline, preventive measures can be implemented to reduce long-term health risks and improve overall well-being.

There were some limitations in the study, including potential dataset biases, computational constraints, and DBSCAN's clustering challenges. The dataset may not fully represent all lifestyle and genetic factors affecting longevity, and the clustering models were limited by the computational resources available. DBSCAN's inability to create well-separated clusters limited its usefulness in this analysis.



Radar Chart for Agglomerative Clustering

Key observations for Agglomerative Clustering:

Cluster 0: Appears to have the lowest overall risk factors, potentially representing a healthier population.

Cluster 1: Has high values in HeartDiseaseorAttack and Stroke, possibly representing older individuals or those with significant cardiovascular risks.

Cluster 2: Has a strong presence in Smoker and HighChol, possibly highlighting lifestyle-driven risks. BMI is also elevated, suggesting a link between obesity and smoking.

Cluster 3: This group stands out in NoDocbcCost (inability to afford healthcare) and Mental Health, indicating financial and psychological struggles.

Cluster 4: High values in BMI, DiffWalk, and PhysHlth, indicating significant physical health challenges. Elevated HighBP and Diabetes_binary, making this a high-risk group.

Future research should consider incorporating additional lifestyle factors such as diet, exercise frequency, and exploring more advanced clustering techniques. Testing different parameter optimizations for DBSCAN or hybrid models combining multiple clustering approaches may also improve clustering performance in health-related studies.

Identifying health risk patterns through clustering allows for better early intervention strategies and public health planning. Understanding how lifestyle choices shape long-term health can lead to more effective prevention efforts, ultimately improving quality of life and reducing the burden of chronic diseases on healthcare systems.

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